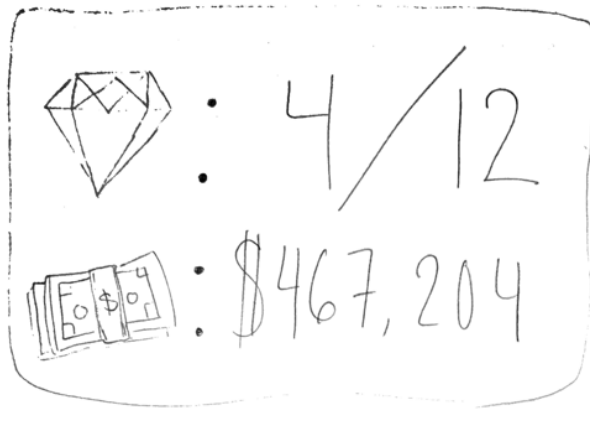


Project 2: Varun Hari, Arda Keskinpala, Sriram Kalle-Marasanapalle, Arlene Shergill

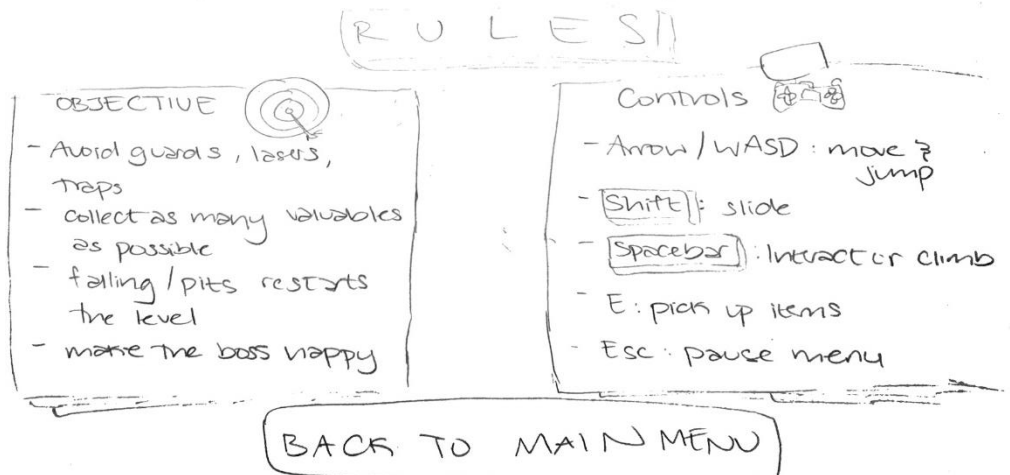
- What do you think the hardest of your assigned mechanics to implement will be?
 - We think that the hardest mechanic to implement will be tactical fights.
 - Unlike normal enemies, tactical fights will require us to create enemies with complicated attack patterns and strategies.
 - This will be difficult because we will have to do a lot of testing
 - In order to balance the difficulty of the fights and there will be a lot more code logic involved in the attacks of the mobs.
- What do you think the easiest of your assigned mechanics to implement will be?
 - We think that the easiest mechanic to implement will be pits
 - We just need think about the y coordinate and maintain a minimum level and time to kill the player.
 - Keep the levels flat
 - Jumping on roof tops for example
- What is your story, theme, and/or inspiration?
 - We were looking at the recent news of the Louvre robbery and felt that a museum heist would be fun to play.
 - We want to have the heist take place in a European country with the main goal of stealing as many collectibles as possible
 - You (the thief) are trying to move up the levels of the museum without triggering lasers, guards, and other mobs.
 - Your final mission is to deliver it to your employer
 - Your employer, a mafia boss, betrays you and is the final boss
- What do you want your players to experience or feel when they are playing your game?
 - We want our players to feel a sense of urgency and secrecy.
 - The whole point of the theme is to do the missions without getting caught, so the player always needs to be moving without stalling for too long.
 - We want the music to be tense as well and match the current actions of the player
- What is your start screen going to look like?
 - We were thinking of doing a picture of the main player (dressed in all black and like a robber) looking at the museum
 - Player can get a sense of the game before playing at first glance
 - There is a button that says “enter” at the center of the screen
 - Another button underneath says rules which the player can press to get the general rules of the game



-
- What is your scoreboard screen going to look like?
 - It's going to always be on the corner of the playing screen. It will showcase the number of collectibles collected by your player out of the total amount the game has to collect (you are not required to collect them all, but the more you do the more \$\$ you get). It will also showcase your dollar value, being the value of your collectibles along with gems/cash you pick up on the way (kind of like coins in Super Mario).



-
- What is your rules screen going to look like?
 - This is a rough draft, but we were thinking of making it look like sticky notes on a dark wall, set down with tapes and pins to give it that heist effect and make it dark and on theme still.



- Are you going to have a tutorial? If so what will it cover?
 - The tutorial will be during game play, and when the situation is right (eg. The player first starts moving and will need to do a jump to keep progressing), slow them down in time, give them instructions w/ the desired button (Press the space bar to jump).
 - This will be done for each level introducing a new mechanism that wasn't in the previous level (since each level will increase in difficulty)

Tutorial will be during the first 30 sec of level 1.

Ex.

freeze



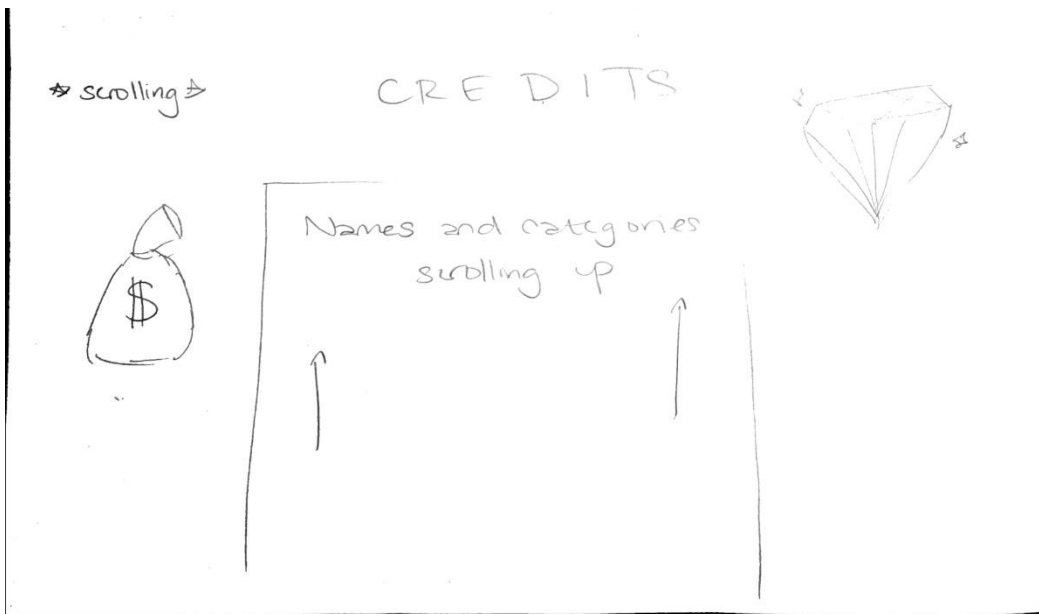
To jump, click
[space] button.



To roll, press
[down arrow] button.



- What is your game credits screen going to look like?
 - It will feature graphics that match the heist theme or maybe sprites of guards/different characters we included on the side while our names/categories/roles scroll up slowly and will be shown as an optional button after the player finishes the boss level.



- How are you going to calculate and retain scores?

- We calculate the final score based on the time that it takes to complete the level as well as the amount of gold stolen and xp gained from killing the mobs.
 - The biggest part of keeping track of scores is separating the different aspects such as having a leaderboard for fastest time and a leaderboard for most mobs killed etc.

First Level: Arlene

- What is your first level going to look like?
 - The first level begins with the player's character entering the Mafia Boss's lair, filled with luxury décor and shadows, with slow and suspenseful music. A short dialogue cutscene introduces the main plot, with the boss assigning the player to heist expensive artifacts from the city's main Museum.
 - Once the dialogue ends, the player gains control of the character and exits through a tall window in the lair, triggering the start of the platform sequence. The scene outside on the balcony shows the nighttime cityscape with glowing streetlights, rooftops, and a bustling city sound.
 - The player must navigate through a series of rooftops, scaffolding, ladders, precise jumps and timing, along with unstable surfaces (crumbling edges, slanted pipes that slip), forcing quick movement.
 - Midway through getting close to the museum, the level branches into two possible routes:
 - Stealth Path: Longer but safer; involves sneaking past patrolling guards, staying hidden behind vents or chimneys, and avoiding motion-detecting lights.
 - Speed Path: Riskier but faster; includes wide gaps, moving platforms, and collapsing rooftops that test agility and timing.
 - Both paths eventually merge near the museum's exterior. The final jump for the player is into a window left open on the top of the museum, marking the end of level 1 and leading them into level 2.
- What mechanics will be introduced in the first level?
 - Horizontal scrolling
 - Scrolling screen
 - One-way collision platforms
 - Moving platforms
 - Ladders
 - Pits

- Unstable/interval platforms
- Rooms:
- Tasks:
- Collectables
- Long jump
- Gliding
- Sliding:
- Health bar/hit points
- How are you going to teach the player about the mechanics they will need to use?
 - This is important given this is the first level in the game (for player mechanics such as jump, roll, sliding, gliding, other mechanics that require an extra keystroke that aren't apparent)
 - At the start of the level (there will be no apparent opponents during this time), the player will do a test run starting at the window of the mafia boss's lair
 - When the situation is right (eg. The player jumps down from the lair window onto a building), slow them down in time, give them instructions w/ the desired button (Press the up arrow to jump)
 - This will be done for each level introducing a new mechanism that wasn't in the previous level (since each level will increase in difficulty)
- What is the goal that must be achieved to indicate you have completed the first level?
 - The player will jump into the museum from the top, and the screen will slowly fade out and transition to the second scene. This will be very obvious and apparent that they finished the level, and the screen when the second scene starts will say "Level 2: The Heist" to ensure understanding.

What is your second level going to look like?

- Our second level is going to be inside the museum
- Our unkillable mobs will be museum guards to ensure no negative messages are spread
- We will have different artifacts come in as the player sprints through
 - These artifacts can be picked up by the player
 - We use these artifacts to help calculate score
- The player will be working in low visibility since the heist is happening in the dark. so, the player can use a flashlight to help see their surroundings and visualize the sprites field of vision

- Aimable weapons (flashlight)
- Tasks

What mechanics will be introduced in the second level?

- The mechanics in this level that are brand new are:
 - A new type of enemy that you shouldn't attack (should be a pacifist with)
 - Collectibles: there being artifacts that can help you fight. Kind of like Night in the Museum, the artifacts can come to life and get superpowers.
 - Powerups that help you in the new dark environment
 - Aimable weapons
 - Armor
 - Rooms
 - Levers and Buttons
 - Tasks

How are you going to teach the player about the mechanics they will need to use?

- Mechanics will be taught by forcing the player to use them in scenarios
 - When the situation is right, slow down the time and give them instruction on the bottom of the screen
 - We need to design the level so that each new mechanic has to be used

How is your second level going to be more challenging or build upon the first level?

- The second level brings in the idea of new NPCs that you need to avoid
- The second level builds on the first level in a thematic sense since you broke in after taking the journey of getting there
- The second level requires you to look out for traps and disable them
- The second level also has time bonuses and requires you to think fast as the screen is constantly moving.

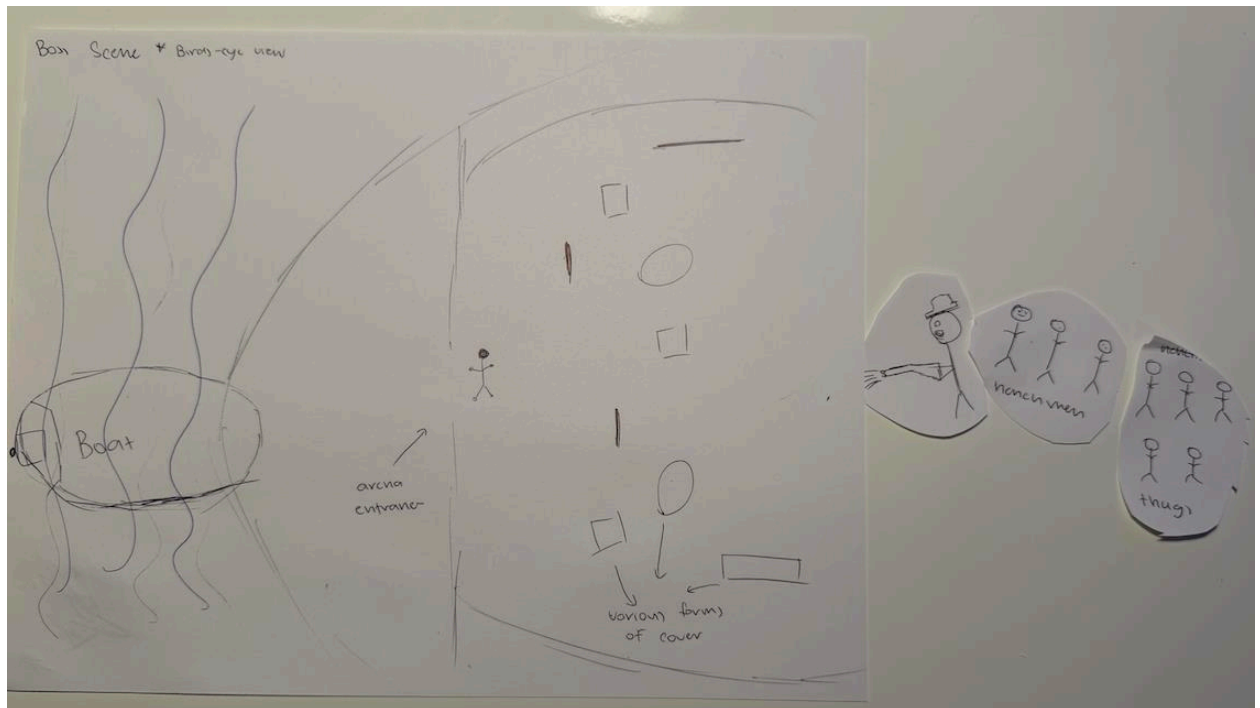
What is the goal that must be achieved to indicate you have completed the second level?

- You need to successfully steal the main artifact (the most expensive one) and make it to the last room in time

- We would have an animation of you breaking out of a window to escape
 - We would hint that they need to break out of a window but if they forget at the very end they need to restart the level
 - This would trigger the next scene
- Third Level: Escaping boss scene
 - What is your third level going to look like?
 - This will be the chase scene where you escape the museum and head to the final boss island
 - The player will first speak with the Boss and decide to escape with the items they stole
 - They will grab a parachute and jump off a building and run to a sewer system
 - The player than move underground to escape notice of the police
 - There will be toxic water and “rats” that deal with damage.
 - What mechanics will be used in your third level?
 - scrolling screens
 - parallax scrolling
 - Ladder
 - Static obstacles
 - Player health
 - Invulnerable enemies
 - Wall Jump
 - What mechanics will be introduced in the third level?
 - New mechanics
 - Gliding
 - Scrolling screens
 - Parallax scrolling
 - How are you going to teach the player about the mechanics they will need to use?
 - The gliding will be intuitive since you will get the parachute gadget and the way you glide will be on screen
 - The chase will also be somewhat intuitive, but when we start the chase out, it will display on screen that they have to use arrow keys to move lanes.

- How is your third level going to be more challenging or build upon the second level?
 - It isn't necessarily more challenging, but it will build upon level two and act as a leadup to level 4
 - This is going to explain a setting change to a remote island where you will fight the final boss.
 - It also adds more depth to the story since it will show the player failing to escape before the fight the boss
 - It won't be comparable in difficulty because the mechanics will be different
- What is the goal that must be achieved to indicate you have completed the third level?
 - The entire scene would change with there being a cut scene where the player needs to get into a sewer system.
 - The surroundings are indicative of being underground (rocks surrounding all 4 sides etc)
 - At around the 2 min mark of that specific attempt of it, they will move on in a cutscene
 - It will be clear level 3 is over since the setting will have left the museum and the mobs get harder to beat
- Boss Level:
 - What is your boss level going to look like?
 - After the sewer escape, you arrive to an abandoned island after a cutscene of you stealing a boat i
 - The setting is like a tropical abandoned military compound
 - The mafia boss is waiting there for you with mercenaries, you think "WHAT, how did you get here so quick???"
 - The arena is like a circular courtyard
 - The arena has scattered crates and pillar for cover
 - It has a central raised platform
 - It has crumbling walls on the perimeter
 - Palm trees in the background
 - The mafia boss appears in a small suit, with bodyguards behind him.
 - He doesn't fight you immediately though, he commands his mercenaries to fight until he runs out of them

- If you get to the point where you can kill his workers, then you can fight him once and for all



- What mechanics will be used in the boss level?
 - Health bar/hit points
 - Long Jump
 - Sliding
 - Slow Projectiles
 - Area of Effect damage
 - Tactical Fights
 - 1-Shot Enemies
 - Multi-Hit Enemies
 - Aimable weapons
 - Multi hit enemies
- What mechanics will be introduced in the boss level?
 - Slow Projectiles
 - Area of Effect Damage
 - Tactical Fights
 - 1-Shot Enemies
 - Multi-Hit Enemies
 - Tactical fights

- How are you going to teach the player about the mechanics they will need to use?
 - The player will get used to mechanics when fighting the mercenaries of the mafia boss, with instructions that show up on the screen for new mechanisms
- How is your boss level going to be differentiated from your regular levels?
 - The boss level is going to have a phase to it, which is not found in other parts of our game
 - The first phase is the mercenaries/thug phase, where you fight the mafias thugs with 1-shot enemies that rush in small groups, and you must kill all of them. They also have special abilities, though.
 - The second phase is 2 bodyguard, which are multi-hit and much stronger than the thugs
 - Finally it's the boss fight, where he has a wide array of attacks that will deal with lots of damage, so be careful!
- What is the goal that must be achieved to indicate you have completed the boss level?
 - The goal is to defeat the mafia boss, and you get to keep the things that you stole, and a helicopter appears, you jump into it, and the screen fades out and then shows you your score!
- What visual assets are you going to need for your game? What are they going to look like?
 - Pixel art for the player character while standing as well as art for the various different actions (running, gliding, sliding, etc.)
 - This includes any assets to help animate small movement gestures to show air moving as you jump or slide through a textured surface.
 - Different levels of surfaces may be needed as well, especially
 - Pixel art will also be needed for the different projectiles in the game. This could be in the form of player weapons as well as bullets from the crooks at level 4
 - Mob art would also be needed for example rats in the sewer, or the bodyguards etc.
 - The surroundings for each of the scenes would also need art

- Spikes in the cave system underground in level three would require different art especially since they act as traps
 - The lasers in scene 2 are moving and can deal damage
- Pixel art for the different enemies we have in our game with different colored clothes or some sort of symbol to identify what they are
- Some sort of clock so that the player can keep track of their time played so far as we can calculate their score with it. This can be displayed in the HUD.
- What sound assets are you going to need for your game? What are they going to sound like?
 - Audio cues when it is dark in the museum, the guards will make footstep sounds that alert you of when they are there
 - Throughout our whole game we will have a song backtrack that accentuates the heist aspect of the game
 - Maybe something similar to mission impossible throughout the whole game that adds that robbery feeling
 - After successfully stealing the money, we will also play like a victory soundtrack
 - Maybe bella caoi? (money heist reference)
 - There would be sound affects for players to give them cues
 - Mobs would give off some sound affects (like “yelling” to signal incoming police officer) or skittering to show upcoming rats
 - There would also be sounds in game for shattering of glass or breaking into the sewer grate to give the game a more realistic feel
 - Breaking any items in the game would also have a shattering sound

Mechanism	Implementation	Dependence on other mechanisms
Scrolling		
Scrolling Screens	- Vertical top-down scrolling screen for boat chase scene. This will force the player to move the boat or else they will hit the obstacles in water. (L3)	Scrolling screens depends on the level the player is on and the rate at which they are moving to determine background speed.

Parallax Scrolling	- While gliding down to the ground and while running to the boat, you will be able to see mountains in the background. - The mountain will move slower than the rest of the scene (L3). - This would depend on the speed of the player	The parallax scrolling depends on the players movement across the screen.
Horizontal Scrolling	- The camera follows the player as they move across rooftops and Platforms (L1). - The player is forced to keep moving forward in order to avoid getting caught by the guards. If they move past the screen they are “caught” and lose the game (L2). - The scrolling depends on the timer in level 2 for the speed.	The scrolling itself depends on the movement
Platforms		
One-Way Collision Platforms	- For rooftop and ledge jumps (L1).	This depends on where the player is facing and how they choose to move.
Moving Platforms	- Appear in the “Speed Path” section for agility jumps (L1).	Moving platforms would depend on tasks. If a player decides to take the speed path, they would meet these platforms.
Ladders	- Used to climb between roof levels and scaffolding (L1). - You use a ladder to go up to the balcony before gliding down (L3).	The ladders depend on the one way collisions platforms to determine where to start and end. It doesn't make sense for

		there to be a floating ladder.
Conveyers	- Without a classic conveyer we can use the wind as a type of loop that helps the player jump further if they run along with it (L1) - If you run against the wind to escape mobs you move a lot slower	Conveyers depend on the scene and the environmental conditions. Adding air acts both as a moving platform that pushes you forward.
Pits	- Representing long drops between buildings falling results in a fail and a restart of the level (L1). Water between the boat and the dock is a pit. (L3)	Pits depend on the one way platforms on the ground to determine where to start and end. You can't have a pit that a player can't clear so make sure to have floating ledges or smaller pits for the player to cross. The pits also relate to gliding and long jumps to cross them.
Interval Platforms	- Crumbling edges on buildings or slipping pipes that require quick movement (L1). - We can have glass cases where you (the thief) can stand on top off to avoid getting caught by the guards, but these cases appear and disappear depending on how close other mobs are (L2).	Depends on pits and the time (in the case of L2)
Progression		
Levers/Buttons	- Can open doors with levers when completing tasks to unlock the thing to activate the lever.	- Depends on tasks to be able to find and use the buttons

	- Each artifact is guarded by lasers, to turn the lasers off, the player must find and press a button to deactivate lasers. To find buttons the player has to depend on the platforms to search around each room before time is up.	- These tasks will lead you to the buttons and levers
Rooms	- The level begins in an indoor setting (the Mafia Boss's lair) and transitions outdoors (L1). - The player must move past certain rooms in the museum and get through them each room that the player travels through marks their progression. Each room will have different artifacts they can try to steal in the time limit, as well as different mobs (L2).	- Rooms depends on buttons and levers to enter - Rooms would also have loot that translates to powerups for the player
Tasks	- Reaching the museum undetected serves as the first task of mission (L1). - Getting identification to open a room that has scanner (L2). To move past the rooms, player has to finish certain tasks that turn off lasers. (L2)	Tasks depend on the lever in L2. The player must find the lever and unlock the door. To do this they must fight other mobs.
Damage To Players		
Slow Projectiles	- Enemies throw grenades with visible trajectories; there is 1-2 second travel time between	It depends on the mob type and the player armor to determine the damage.

	thrown and landing so that player has time to react and get away from its radius (L4).	
Area of Effect Damage	- A red circular warning appears on the ground after grenade explosion; the circle shows the damage radius that ticks your health when inside of it, can force a player to move (L4).	The area of damage depends on the type of projectile. Bullets and arrows would be a small area while a grenade in L4 would have a larger area.
Static Obstacles	- These would just be natural barriers that we could create to simulate (L2). - Different static obstacles on land that needed to be jumped over and then on sea like sharks, buoys, logs that can't be hit to continue (L3).	These would depend on weapon attacks to break
Intermittent Obstacles	- In level 3, the boat rides have logs in the path that you have randomly have to dodge in the water	This doesnt depend on other things other than the level and perhaps the room to change the difficult and summoning speed
Pattern Attacks	- The boss (l4) would have attacks that the player could learn. - This would be similar to how sonic attacks work when trying to attack doctor egg head on his space machine. - When the boss is at half health it starts letting out certain patterns of powerful attacks like beams from the ground etc.	Depends on the level and the room to change how often. - This would also depend on

Meta Gameplay Goals		
Collectables	- Optional small items (jewels or cash) scattered on rooftops (almost like coins), will be added to player scoreboards “value” (General). - These artifacts can give them temporary power ups or will go towards calculating the entire score.	These would be dropped by mobs and things you can pick up as you go on
Advanced Player Movement		
Long Jump	- Required to cross large gaps between buildings (L1). - Used to leap between cover spots in the arena, essential for dodging attacks from stronger enemies in the boss fight (L4).	This depends more on pits as certain pits would be larger
Wall Jump	- Can wall jump off of certain walls in the museum when trying to reach special rooms to complete tasks (L3)	The wall jump relies on the type of surface that the player is on
Sliding	- Used when escaping guards or slipping down angled pipes (L1,L3). - Allows players to quickly duck under slow projectiles or quickly get behind cover (L4).	This depends on incline surfaces that are stationary
Gliding	- Player glides from the top of the building to the ground level using a parachute (L3).	This depends on pits and large jumps
Power Ups		

Power Ups (Pickup Items)	- As mentioned before the powerups will be the collectables and will give different temporary buffs - o Examples are: lighting the dark room up - o Giving an extra life - o Acting as shields for the lasers	This depends on the tasks and how fast you solve them
Player Health		
Health Bar/Hit Points	- Player health will start off at the amount it was on at the end of the last level. You can increase health by meeting hidden guidelines like finishing the level in a predetermined time. - o You lose health by getting attacked by mobs. If your health hits zero you die and have to restart the entire game (rogue like)(General).	This depends on damage you take from mobs
Armor	- Armor would just be in the form of having an extra life (acting as the shield you need) (this would scenes 2-4)	This is a upgradeable stat.
Player Attacks		
Aimable Weapons	- The player has a flashlight they they can use as a weapon against ghosts as normal weapons don't work on them - Keeping the flashlight aimed on the ghost using	This depends on the upgradeable weapons as your weapon gets better, you can get more range

	the curser can cause them to take damage - Ghosts are mobs that are multi hit but can surprise attack you in the dark (L2)	
Upgradeable Weapons	- The player has only a few types of weapons - But each level, using the power up pick-ups they can invest certain cells into making their weapons stronger - Kind of like a trade off - This is in all 4 levels	This depends on the xp you collect from the mobs you kill
Pacifist	- The player can't attack in this level as they don't want to trigger any alarms. - Attacking an NPC results in you losing a heart so it can be strategic - This affects player health	This depends on the players health and the mobs health
Enemy Types		
1-Shot Enemies	- Weak thugs that die from one hit, they spawn in big groups though and move at faster speeds than stronger mobs (L4).	This depends on the health of the mob
Multi-hit Enemies	- Bodyguards that require more hits to defeat are stronger, but slower. These are part of the second phase of the boss battle (L4). - How many hits depends on how upgraded your weapon is	This depends on the type of mob and the health
Tactical Fights	- The boss has 3 distinct phases with different attack patterns. The player must learn how to	The tactical fight depends on mob type

	adapt to each pattern to get through the boss level (L4).	
Invulnerable Enemies	- Museum guards that you can stun when looking at them, to give yourself more time (like Pac man) (L2). - You cannot hurt any of the enemies that chase you, but they can hurt you. They will move at a constant speed so it will be possible to avoid them (L3).	This depends on the level the player is on to determine the mob. The knockback depends on armor the player has